Homotopy types of schemes

Down under, August 2026

Goal of lecture series. Give a "modern" introduction to étale homotopy theory. Explain recent work refining the étale homotopy type using condensed math.

Some References. Most of what we'll cover is contained in:

> Hoyois, "Higher Galois theory"

> H-Holzschuh-Wolf, "The fundamental fiber sequence in étale homotopy theory"

> H-Holzschuh-Wolf, "Nonabelian basechange theorems & étale homotopy theory"

> Barwick-Glasman-H, "Exodromy"

> H-Holzschuh-Lara-Mair-Martini-Wolf, "The condensed homotopy type of a scheme"

Lecture 1

Notation An = ∞-category of anima / spaces / ∞-groupoids

$|-|: \mathrm{Top} \longrightarrow \mathrm{An}$ underlying anima / homotopy type

Desire of étale homotopy theory. Construct an étale homotopy type

$$\Pi_\infty^{\text{ét}} : \mathrm{Sch} \longrightarrow \mathrm{Pro}(\mathrm{An})$$

such that

$$\pi_1 \Pi_\infty^{\text{ét}}(X) \simeq \pi_1^{\text{ét}}(X)$$

there are two variants; one is a pro-group, the other is profinite

$$H^*(\Pi_\infty^{\text{ét}}(X), R) \simeq H^*_{\text{ét}}(X; R)$$

discrete ring

and $\Pi_\infty^{\text{ét}}(X)$ "satisfies all of the basic theorems in étale cohomology"

Problems

(1) $\pi_1^{\text{ét}}(X)$ is a pro-object

(2) In algebraic geometry, we don't have a reasonable space to do homotopy theory with – we have an ∞-category of sheaves.

Question that will motivate our definition. Given a reasonable topological space T, how can we recover the underlying anima $|T|$ from the ∞-category $Sh(T)$ of sheaves of anima on T?

Shape theory in topology

AKA The nonabelian version of comparing singular and sheaf cohomology

Technical Digression (on hypersheaves). Let T be a space. Consider the following full subcategories of

$$\mathrm{PSh}(T) := \mathrm{Fun}(\mathrm{Open}(T)^{op}, \mathrm{An})$$

(1) The full subcategory $\mathrm{Sh}(T)$ spanned by the sheaves

(2) The full subcategory $\mathrm{Sh}^{hyp}(T)$ spanned by the presheaves that satisfy descent for hypercoverings

(3) The localization of $\mathrm{PSh}(T)$ at the stalk-wise equivalences.

Then $(2) = (3)$, and $\mathrm{Sh}^{hyp}(T) \subset \mathrm{Sh}(T)$ admits a left exact left adjoint. However, the inclusion is generally strict

Ex If T admits a CW structure, then $\mathrm{Sh}^{hyp}(T) = \mathrm{Sh}(T)$.

Basis Lemma If $\mathcal{B} \subset \mathrm{Open}(T)$ is a basis for the topology on T, then the restriction functor

$$\mathrm{Sh}^{hyp}(T) \longrightarrow \mathrm{Sh}^{hyp}(\mathcal{B})$$

is an equivalence with inverse right Kan extension along the inclusion $\mathcal{B}^{op} \subset \mathrm{Open}(T)^{op}$

van Kampen Theorem The functor $|\text{-}| : \mathrm{Top} \longrightarrow \mathrm{An}$ is a hypercomplete cosheaf, i.e., for every space T and hypercovering $U_\bullet \longrightarrow T$, the natural map

$$\operatorname*{Colim}_{[n] \in \Delta^{op}} |U_n| \longrightarrow |T|$$

is an equivalence.

Construction Let T be a space. Write $L : \mathrm{PSh}(T) \longrightarrow \mathrm{An}$ for the left Kan extension of $|-| : \mathrm{Open}(T) \longrightarrow \mathrm{An}$ along the Yoneda embedding:

$$\begin{tikzcd} \mathrm{Open}(T) \arrow[r, "|-|"] \arrow[d, hook] & \mathrm{An} \\ \mathrm{PSh}(T) \arrow[ur, "L"'] & \end{tikzcd}$$

Then L has a right adjoint R given by the formula

$$R(A)(U) = \mathrm{Map}_{\mathrm{An}}(|U|, A).$$

> By the van Kampen theorem, R factors through $\mathrm{Sh}^{\mathrm{hyp}}(T)$.

Prop If T is a locally weakly contractible space, then the functor $L : \mathrm{Sh}^{\mathrm{hyp}}(T) \longrightarrow \mathrm{An}$ is left adjoint to the constant hypersheaf functor $T^* : \mathrm{An} \longrightarrow \mathrm{Sh}^{\mathrm{hyp}}(T)$.

Proof.
Write $\mathrm{Open}_{wc}(T) \subset \mathrm{Open}(T)$ for the subposet of weakly contractible opens. Notice that the composite

basis Lem

$$\mathrm{An} \xrightarrow{R} \mathrm{Sh}^{hyp}(T) \underset{\mathrm{rest}}{\xrightarrow{\sim}} \mathrm{Sh}^{hyp}(\mathrm{Open}_{wc}(T))$$

$$A \longmapsto [U \mapsto \mathrm{Map}(\underbrace{|U|}_{*}, A) \simeq A]$$

is the constant presheaf functor. But it's also a sheaf! $\square$

Remark. This implies that for a locally weakly contractible space T and abelian group M,

$$C^{-*}_{\mathrm{Sing}}(T; M) \simeq R\Gamma_{\mathrm{sheaf}}(T; M).$$

And the argument is easy!

Upshot We can recover $|T|$ by applying the left adjoint to the const. hypersheaf functor to the terminal object.

Idea. In algebraic geometry, we should use the same definition, replacing $\mathrm{Sh}^{\mathrm{hyp}}(T)$ with the ∞-topos of étale sheaves on a scheme.

⚠ The constant étale sheaf functor doesn't admit a left adjoint. But it admits a **pro-left adjoint**.

Crash course on ∞-topoi

Thm. Let $\mathcal{X}$ be an ∞-category. TFAE:

(1) ∃ a small ∞-cat $\mathcal{C}$ and a left exact accessible localization

$$\begin{tikzcd} \mathcal{X} \arrow[r, hook, shift right, "\text{acces.}"'] & \mathrm{PSh}(\mathcal{C}) \arrow[l, shift right, "\text{lex}"'] \arrow[l, phantom, "\perp"] \end{tikzcd}$$

(2) Giraud's axioms: $\mathcal{X}$ is presentable and

(a) Coproducts are disjoint: $\forall\, U, V \in \mathcal{X}$, the square

$$\begin{tikzcd} \emptyset \arrow[r] \arrow[d] & V \arrow[d] \\ U \arrow[r] & U \sqcup V \end{tikzcd}$$

is a pullback

(b) Colimits are universal: for every diagram $U_\bullet : A \to \mathcal{X}$, objects $V, W \in \mathcal{X}$ and morphisms

$$\operatorname*{colim}_{\alpha \in A} U_\alpha \longrightarrow W \longleftarrow V$$

we have

$$\operatorname*{colim}_{\alpha \in A} \left(U_\alpha \times_W V\right) \xrightarrow{\sim} \left(\operatorname*{colim}_{\alpha \in A} U_\alpha\right) \times_W V.$$

(c) Groupoid objects are effective: if $U_\bullet : \Delta^{op} \to \mathcal{X}$ is a simplicial object so that for every partition $[n] = S \cup T$ with $S \cap T = \{i\}$ a single point, then

$$\begin{tikzcd} U_n \simeq U_\bullet([n]) \arrow[r] \arrow[d] & U_\bullet(T) \arrow[d] \\ U_\bullet(S) \arrow[r] & U_\bullet(\{i\}) \simeq U_0 \end{tikzcd}$$

is a pullback,

then $U_\bullet \simeq \underbrace{\check{C}\text{ech nerve of } U_0 \to |U_\bullet|}$

$f: X \to Y$ Čech nerve is $\cdots \; X \times_Y X \underset{pr_2}{\overset{pr_1}{\rightrightarrows}} X$

(3) $\mathcal{X}$ is presentable and colimits in $\mathcal{X}$ are **van Kampen**: the functor

$$\begin{tikzcd} \mathcal{X}^{op} \arrow[r] & \mathrm{Cat}_\infty^{large} \\ U \arrow[r, mapsto] \arrow[d] & \mathcal{X}_{/U} \\ V \arrow[r, mapsto] & \mathcal{X}_{/V} \arrow[u, "U \times_V (-)"'] \end{tikzcd}$$

preserves limits, i.e., $\mathcal{X}_{/\operatorname{colim}_{\alpha \in A} U_\alpha} \xrightarrow{\sim} \lim_{\alpha \in A^{op}} \mathcal{X}_{/U_\alpha}$.

⚠ Note that we're not saying that there is an ∞-site $(\mathcal{C}, \tau)$ and an equivalence $\mathcal{X} \simeq Sh_\tau(\mathcal{C})$.
This is expected to be false!

Cor (of (3)). If $\mathcal{X}$ is an ∞-topos, then for any $U \in \mathcal{X}$, the ∞-cat $\mathcal{X}_{/U}$ is an ∞-topos.

Ex. T space, $U \subset T$ open, $h(U) \in Sh(T)$ sheaf represented by U. Then

$$Sh(T)_{/h(U)} \simeq Sh(U).$$

Ntn. For a scheme X, we write $X_{ét} = Sh(Ét_X; An)$ for the étale ∞-topos of X.

Ex. X scheme, $U \in Ét_X$, $h(U) \in X_{ét}$ sheaf represented by U. Then

$$(X_{ét})_{/h(U)} \simeq U_{ét}.$$

Upshot For a covering $U_\bullet \to T$ or étale covering $U_\bullet \to X$ we have

$$\mathrm{Sh}(T) \xrightarrow{\sim} \lim_{[n]\in\Delta} \mathrm{Sh}(U_n) \quad \text{and} \quad X_{\text{ét}} \xrightarrow{\sim} \lim_{[n]\in\Delta} (U_n)_{\text{ét}}.$$

Def. Given ∞-topoi $\mathcal{X}$ and $\mathcal{Y}$, a **geometric morphism** $f_*: \mathcal{X} \to \mathcal{Y}$ is a right adjoint whose left adjoint $f^*: \mathcal{Y} \to \mathcal{X}$ is left exact (= preserves finite limits).

- $\mathrm{RTop}_\infty := \infty$-topoi and geometric morphisms
- $\mathrm{LTop}_\infty := \infty$-topoi and left exact left adjoints

$$\mathrm{LTop}_\infty \simeq (\mathrm{RTop}_\infty)^{op}$$

Thm [HTT, Prop. 6.3.2.3 & Cor. 6.3.4.7] LTop_∞ has all limits and colimits. Moreover, the forgetful functor

$$\mathrm{LTop}_\infty \longrightarrow \mathrm{Cat}_\infty^{\mathrm{large}}$$

preserves limits.

Lecture 2

Shape Theory

Ntn $\mathcal{X}$ ∞-topos

$\Gamma_* = \mathrm{Map}_{\mathcal{X}}(1_{\mathcal{X}}, -) : \mathcal{X} \longrightarrow \mathrm{An}$ **global sections** functor

with left exact left adjoint $\Gamma^* : \mathrm{An} \longrightarrow \mathcal{X}$.

Recollection Let $\mathcal{C}$ be an ∞-category. The ∞-category $\mathrm{Pro}(\mathcal{C})$ of **pro-objects** in $\mathcal{C}$ is an ∞-category equipped with a functor $y : \mathcal{C} \hookrightarrow \mathrm{Pro}(\mathcal{C})$ satisfying the following universal property

(1) $\mathrm{Pro}(\mathcal{C})$ admits cofiltered limits

(2) If D is an ∞-category with cofiltered limits, the restriction functor

$$\mathrm{Fun}^{\mathrm{cofil}}(\mathrm{Pro}(\mathcal{C}), D) \xrightarrow{- \circ y} \mathrm{Fun}(\mathcal{C}, D)$$

cofiltered limit-preserving

is an equivalence

If $\mathcal{C}$ has finite limits, then $\mathrm{Pro}(\mathcal{C})$ has all limits, and y preserves finite limits.

Observe. For an ∞-topos $\mathcal{X}$, since Γ^* is left exact, the unique cofiltered limit-preserving extension

$$\mathrm{Pro}(\mathrm{An}) \longrightarrow \mathcal{X}$$

of Γ^* preserves all limits. Hence by (an appropriate version of the) adjoint functor theorem, this functor admits a left adjoint

$$\Gamma_\# : \mathcal{X} \longrightarrow \mathrm{Pro}(\mathrm{An})$$

Def.

(1) The **shape** of an ∞-topos $\mathcal{X}$ is

$$\Pi_\infty(\mathcal{X}) = \Gamma_\#(1_{\mathcal{X}})$$

(2) For a scheme X, the **étale homotopy type** of X is

$$\Pi_\infty^{\text{ét}}(X) := \Pi_\infty(X_{\text{ét}}).$$

Ex. If T is a locally weakly contractible space, then

$$\Pi_\infty(\mathrm{Sh}^{\mathrm{hyp}}(T)) \simeq |T|$$

Ex. If T is the Warsaw circle, then $\Pi_\infty(\mathrm{Sh}(T)) \simeq S^1$ as a constant pro-anima

Remark There is a classical theory of shapes of compact Hausdorff spaces. The shape of T in the classical sense coincides with $\Pi_\infty(\mathrm{Sh}(T))$

Ex. If $\mathcal{C}$ is a small ∞-category, then the constant functor $\Gamma^*: \mathrm{An} \longrightarrow \mathrm{Fun}(\mathcal{C}, \mathrm{An})$ admits a genuine left adjoint $\mathrm{colim}_{\mathcal{C}}$. Hence

$$\Pi_\infty(\mathrm{Fun}(\mathcal{C}, \mathrm{An})) \simeq \operatorname{colim}_{\mathcal{C}} *$$

$$\simeq B\mathcal{C}$$

Classifying anima
$B: \mathrm{Cat}_\infty \longrightarrow \mathrm{An}$ left adj to inclusion.

Q. What is the meaning of the shape?

Obs. For any anima A

$$\mathrm{Map}_{\mathrm{Pro(An)}}(\Gamma_{\#}(1_X), A) \simeq \mathrm{Map}_X(1_X, \Gamma^*(A))$$

$$\simeq \Gamma_* \Gamma^*(A)$$ global sections of constant sheaf w/ coefficients in A

Obs. In $\mathrm{Pro(An)}$

$$\mathrm{Map}(\text{"}\lim_i\text{"} A_i, \text{"}\lim_j\text{"} B_j) \simeq \mathrm{colim}_j \mathrm{Map}(\text{"}\lim_i\text{"} A_i, B_j),$$

so a proanima is determined by mapping to all anima.

Lem. The functor

left exact

$$\begin{aligned} \mathrm{Pro(An)} &\xrightarrow{\sim} \mathrm{Fun}^{\mathrm{lex,accessible}}(\mathrm{An}, \mathrm{An})^{\mathrm{op}} \\ A &\longmapsto \mathrm{Map}_{\mathrm{Pro(An)}}(A, -) \end{aligned}$$

is an equivalence of ∞-categories

Upshot. $\Pi_\infty(X) = \Gamma_* \Gamma^* \in \mathrm{Fun}^{\mathrm{lex,acc}}(\mathrm{An}, \mathrm{An})^{\mathrm{op}} \simeq \mathrm{Pro(An)}$

Functonality of the Shape

Obs. The functor

$$\begin{aligned} An &\longrightarrow RTop_\infty \\ A &\longmapsto Fun(A, An) \simeq An_{/A} \end{aligned}$$

w/ right Kan extension functoriality

is left exact and fully faithful. Hence extends to a right adjoint

$$\lambda : Pro(An) \longrightarrow RTop_\infty$$

Thm (Lurie). The left adjoint to λ is on objects given by the shape.

Corollary. For any ∞-topos X and diagram $U_\bullet : I \longrightarrow X$,

$$\operatorname{colim}_{i \in I} \Pi_\infty(X_{/U_i}) \xrightarrow{\sim} \Pi_\infty(X_{/\operatorname{colim}_i U_i})$$

Cor. The étale homotopy type

$$\Pi_\infty^{\text{ét}} : \mathrm{Sch} \longrightarrow \mathrm{Pro}(\mathrm{An})$$

is an étale cosheaf.

Locally constant objects & monodromy

Def Let $\mathcal{X}$ be an ∞-topos. An object $L \in \mathcal{X}$ is **locally constant** if there exists an effective epimorphism $\coprod_{i \in I} U_i \twoheadrightarrow L$ and anima $(A_i)_{i \in I}$ such that there exist equivalences

$$L \times U_i \simeq \Gamma^*_{\mathcal{X}_{/U_i}}(A_i) \qquad \text{in } \mathcal{X}_{/A_i}$$

$$\simeq \Gamma^*_{\mathcal{X}}(A) \times U_i$$

> We say that L is **lisse** if, in addition I can be chosen to be finite and each A_i can be chosen to be π-finite.

$\pi_0(A)$ is finite, $\pi_i(A) = 0$ for $i \gg 0$ and all $\pi_i(A)$ are finite

> We write $\mathcal{X}^{\text{lisse}} \subset \mathcal{X}^{lc} \subset \mathcal{X}$ for the full subcategories of lisse and locally constant objects

Monodromy V1 (Lurie) Let $\mathcal{X}$ be an ∞-topos. If $\Gamma^*: \mathrm{An} \to \mathcal{X}$ admits a genuine left adjoint $\Gamma_\#: \mathcal{X} \longrightarrow \mathrm{An}$, then the right adjoint to

$$\Gamma_\#: \mathcal{X} \longrightarrow \mathrm{An}_{/\Gamma_\#(1_\mathcal{X})} \simeq \mathrm{Fun}(\Pi_\infty(\mathcal{X}), \mathrm{An})$$

Straightening-unstraightening $\mathrm{An}_{/A} \simeq \mathrm{Fun}(A, \mathrm{An})$ + by def. $\Pi_\infty(\mathcal{X}) = \Gamma_\#(1_\mathcal{X})$

is fully faithful with image $\mathcal{X}^{lc}$. That is,

$$\mathcal{X}^{lc} \simeq \mathrm{Fun}(\Pi_\infty(\mathcal{X}), \mathrm{An}).$$

Ex If T is a locally weakly contractible space, then

$$LC^{hyp}(T) \simeq \mathrm{Fun}(|T|, \mathrm{An})$$

> For lisse objects, we don't need any assumptions on $\mathcal{X}$. We first need some technical digressions.

Background on proanima

Obs. The adjunctions $\begin{tikzcd} \mathrm{An} \arrow[r, shift left, "\tau_{\le n}"] \arrow[r, phantom, "\perp"] & \mathrm{An}_{\le n} \arrow[l, shift left] \end{tikzcd}$ induce adjunctions

$$\begin{tikzcd} \mathrm{Pro}(\mathrm{An}) \arrow[r, shift left, "\tau_{\le n}"] \arrow[r, phantom, "\perp"] & \mathrm{Pro}(\mathrm{An}_{\le n}) \arrow[l, shift left] \end{tikzcd}$$

n-truncated anima: $\pi_i(A) = 0$ for $i > n$

⚠ It is possible that a map of proanima $f : X \to Y$ induces an equivalence on all $\tau_{\le n}$, but isn't an equivalence.

If $X \in \mathrm{An}$ regarded as a constant proanima and

$$\tau_{<\infty} A = \{ \cdots \longrightarrow \tau_{\le n+1} A \longrightarrow \tau_{\le n} A \longrightarrow \cdots \},$$

there is a natural map $X \longrightarrow \tau_{<\infty} X$ that's an equiv on all $\tau_{\le n}$. However,

$$\mathrm{Map}_{\mathrm{Pro}(\mathrm{An})}(\tau_{<\infty} A, A) \simeq \operatorname*{Colim}_{n \to \infty} \mathrm{Map}(\tau_{\le n} A, A).$$

So every map $\tau_{<\infty} A \longrightarrow A$ factors through some $\tau_{\le n}$.
So $A \to \tau_{<\infty} A$ can't have an inverse if A isn't n-trunc. for some n.

Fact The inclusion $\mathrm{Pro}(\mathrm{An}_{<\infty}) \hookrightarrow \mathrm{Pro}(\mathrm{An})$ admits a left adjoint $\tau_{<\infty}$ that identifies

$$\mathrm{Pro}(\mathrm{An})\left[\begin{pmatrix}\text{equivs on} \\ \text{all } \tau_{\leq n}\end{pmatrix}^{-1}\right] \xrightarrow{\sim} \mathrm{Pro}(\mathrm{An}_{<\infty})$$

> We call $\tau_{<\infty}$ the **protruncation** functor

Def The ∞-category of **profinite anima** is $\mathrm{Pro}(\mathrm{An}_\pi)$. The inclusion $\mathrm{Pro}(\mathrm{An}_\pi) \hookrightarrow \mathrm{Pro}(\mathrm{An})$ admits a left adjoint $X \mapsto X_\pi^\wedge$ called **profinite completion**

Facts

(1) If S is a set, then $S^{\wedge}_{\pi} \simeq \beta(S)$ Čech-Stone compactification

- Since β doesn't preserve finite products, $(-)^{\wedge}_{\pi}$ generally doesn't either

(2) If A is a connected anima, then

$$\pi_1(A^{\wedge}_{\pi}) \cong \widehat{\pi_1(A)}$$

profinite completion of groups

(3) In general, there exist discrete groups G for which $(BG)^{\wedge}_{\pi}$ has higher homotopy. In general,

$$(BG)^{\wedge}_{\pi} \longrightarrow B\hat{G}$$

is an equivalence if and only if G is **good** in the sense of Serre: for every finite G-module M, the natural map

$$H^*(\hat{G}, M) \longrightarrow H^*(G, M)$$

is an isomorphism.

Monodromy for lisse objects

Ntn Given an ∞-category D, write

$$\mathrm{Fun}^{\mathrm{cts}}(-, D) : \mathrm{Pro}(\mathrm{Cat}_\infty)^{op} \longrightarrow \mathrm{Cat}_\infty$$

for the unique colimit-preserving extension of $\mathrm{Fun}(-, D) : \mathrm{Cat}_\infty^{op} \longrightarrow \mathrm{Cat}_\infty$:

$$\mathrm{Fun}^{\mathrm{cts}}(\text{"}\lim_{i \in I}\text{"} C_i, D) = \operatorname*{colim}_{\in I^{op}} \mathrm{Fun}(C_i, D)$$

Monodromy Thm $\mathcal{X}$ ∞-topos.
There is a natural equivalence

$$\mathcal{X}^{\mathrm{lisse}} \simeq \mathrm{Fun}^{\mathrm{cts}}(\Pi_\infty(\mathcal{X})^\wedge_\pi, \mathrm{An}_\pi).$$

Slogan "The profinite shape classifies lisse local systems"

> In fact

Thm (Lurie). $\lambda : \mathrm{Pro}(\mathrm{An}_\pi) \longrightarrow \mathrm{RTop}_\infty$ is fully faithful with explicit image.

Basic properties of the étale homotopy type

Convention We'll always protruncate and just write $\Pi_\infty^{\acute{e}t}$ for $\tau_{<\infty}\Pi_\infty^{\acute{e}t} = \Pi_{<\infty}^{\acute{e}t}$

Ex. If k is a field, then a choice of separable closure $\bar{k} > k$ gives an equivalence

$$\Pi_\infty^{\acute{e}t}(\mathrm{Spec}(k)) \xrightarrow{\sim} B\mathrm{Gal}(\bar{k}/k)$$

Ex. $\Pi_\infty^{\acute{e}t}(\mathrm{Spec}(\mathbb{Z})) \simeq *$ Spec($\mathbb{Z}$) has no unramified étale covers $\Rightarrow \pi_1^{\acute{e}t} = 0$
Arithmetic duality $\Rightarrow H^{\geq 1}_{\acute{e}t}(\mathrm{Spec}(\mathbb{Z}); \mathbb{Z}/n) = 0$
Hurewicz completes the proof

Thm (profiniteness). If X is a noetherian [geometrically unibranch] scheme, then $\Pi_\infty^{\acute{e}t}(X)$ is profinite (e.g. smooth / field

Fundamental Fiber Sequence Let X/k be a qcqs scheme over a field and $\bar{k} \supset k$ a separable closure. There are natural fiber sequences

$$\Pi_\infty^{\text{ét}}(X_{\bar{k}}) \longrightarrow \Pi_\infty^{\text{ét}}(X) \longrightarrow B\mathrm{Gal}(\bar{k}/k)$$

and

$$\hat{\Pi}_\infty^{\text{ét}}(X_{\bar{k}}) \longrightarrow \hat{\Pi}_\infty^{\text{ét}}(X) \longrightarrow B\mathrm{Gal}(\bar{k}/k$$

Riemann Existence $X/\mathbb{C}$ finite type. There is a natural map

$$|X(\mathbb{C})| \longrightarrow \Pi_\infty^{\text{ét}}(X)$$

induced by a geom. mor $Sh(X(\mathbb{C})) \longrightarrow X_{\text{ét}}$

that becomes an equivalence after profinite completion

Künneth formulas. k field with absolute Galois group G_k, $X, Y/k$ qcqs

(1) If Y is proper$/k$,

$$\hat{\Pi}^{\text{ét}}_{\infty}(X \times_k Y) \xrightarrow{\sim} \hat{\Pi}^{\text{ét}}_{\infty}(X) \times_{BG_k} \hat{\Pi}^{\text{ét}}_{\infty}(Y)$$

(2) $p = \mathrm{char}(k)$. If G_k is prime-to-p, then

$(-)^{\wedge}_{p'}$ completion away from p

$$\Pi^{\text{ét}}_{\infty}(X \times_k Y)^{\wedge}_{p'} \xrightarrow{\sim} \Pi^{\text{ét}}_{\infty}(X)^{\wedge}_{p'} \times_{BG_k} \Pi^{\text{ét}}_{\infty}(Y)^{\wedge}_{p'}$$

Application: Anabelian geometry

Thm (A. Schmidt – Stix). $k \supset \mathbb{Q}$ finitely generated extension, X, Y smooth k-varieties st $\exists$ embeddings $X, Y \xhookrightarrow[\text{loc closed}]{} \prod$ hyperbolic curves

The the natural map

$$\mathrm{Isom}_k(X,Y) \longrightarrow \pi_0 \mathrm{Equiv}_{BG_k}(\Pi_\infty^{ét}(X), \Pi_\infty^{ét}(Y))$$

is a split injection with a natural retraction

Cor. If $\Pi_\infty^{ét}(X) \simeq \Pi_\infty^{ét}(Y)$ over BG_k, then $X \simeq Y$ as k-schemes

Thm (Holschbach – J. Schmidt – Stix) k alg closed field $\mathrm{char}(k) > 0$, X/k smooth k-variety

Then $\Pi_\infty^{ét}(X) \simeq *$ if and only if $X \simeq \mathrm{Spec}(k)$.

Thm (Achinger 2017) If X is an affine connected $\mathbb{F}_p$-scheme, then $\hat{\Pi}^{\text{ét}}_{\infty}(X)$ is 1-truncated.

See Holzschuh's talk for more applications

Application: The Artin-Tate Pairing

Setup. $X/\mathbb{F}_q$ smooth geometrically connected surface   dim $= 2d$

$\ell \nmid q$ prime $\mathrm{Br}_{nd}(X) := \dfrac{\mathrm{Br}(X)}{\text{divisible elts}}$

> M. Artin & Tate defined a pairing

$$\mathrm{Br}_{nd}(X)[\ell^\infty] \times \mathrm{Br}_{nd}(X)[\ell^\infty] \longrightarrow \mathbb{Q}/\mathbb{Z}.$$

Really defined on $H^{2d}_{ét}(X; \mathbb{Q}_\ell/\mathbb{Z}_\ell(d))_{nd} \xrightarrow{\sim} H^{2d+1}_{ét}(X; \mathbb{Z}_\ell(d))_{tors}$

Conj (Tate, 1966) The Artin-Tate pairing is alternating

History

1967 & 86 Manin gave counterexamples

1996 Urabe found mistakes is Manin's work

Thm (Feng, 2017) Tate's conjecture is true

> uses analogies from arithmetic topology

Idea. Because of the fiber sequence

$$\Pi_\infty^{\acute{e}t}(X_{\overline{\mathbb{F}}_q}) \longrightarrow \Pi_\infty^{\acute{e}t}(X) \longrightarrow B\hat{\mathbb{Z}} \simeq (S^1)^\wedge_\pi$$

"$\mathbb{R}$-dim" "dim 4" "dim 5" "dim 1"

$\Pi_\infty^{\acute{e}t}(X)$ behaves like a real 5-manifold

Artin-Tate pairing $\longleftrightarrow$ linking form on an orientable $(4d+1)$-manifold

> Feng used the étale homotopy type to construct Steenrod operations on étale cohomology + analogies from topology to prove the conjecture